EXCLUSION LAWS FOR CONSECUTIVE ARTIN PRIMES IN ARBITRARY BASES

JOSH BALD

ABSTRACT. Say that a prime p is *Artin base a* if a is a primitive root modulo p. We determine, for every non-square base a, the exact set of gap classes at which two primes can *never* both be Artin base a. Two distinct mechanisms produce such *exclusion classes*. The first is *reversal*: the quadratic character χ_a satisfies $\chi_a(q) = -\chi_a(p)$ for all admissible $p \equiv q - g$, so one of the two primes is a quadratic residue base a and hence not a primitive root; we classify the reversing classes completely in terms of the factorization of the discriminant of $\mathbb{Q}(\sqrt{a})$ into prime discriminants. The second, which operates even when no reversing class exists, is *inadmissibility*: for $d = 5$ (uniquely among squarefree parts $d \equiv 1 \bmod 4$ coprime to 6) the residue classes carrying $\chi_5 = -1$ are so sparse that for $g \equiv 2, 3 \pmod 5$ every consecutive configuration with both characters -1 would force $5 \mid p$ or $5 \mid q$; we prove an exact count $\#\{r \bmod d : \chi(r) = \chi(r + g) = -1\} = (d - 3 + 2\chi(g))/4$ for prime $d \equiv 1 \bmod 4$, which vanishes precisely at $d = 5$, $\chi(g) = -1$. The resulting complete dichotomy is: base a admits an exclusion law if and only if d is even, $d \equiv 3 \pmod 4$, $3 \mid d$, or $d = 5$; the smallest bases with no exclusion law are $d = 13, 17, 29$. The classification is verified by exhaustive computation for all non-square $2 \le a \le 80$, and the exclusion laws are verified on 84,981,870 consecutive prime pairs below 10^9 across eight bases, with zero exceptions.

We then measure, for eleven bases, the consecutive-pair Artin correlation $\delta(a)$ over all 50,847,531 consecutive prime pairs below 10^9. Here the exclusion weight w (the fraction of consecutive pairs whose gap lies in an exclusion class, now including the inadmissibility classes of base 5) correlates with $|\delta|$ at $r = 0.65$ — but this association is confounded with the conductor: $r(\log f, |\delta|) = -0.96$, the partial correlation of w given $\log f$ is only 0.14, and the three bases with no exclusion classes at all $(13, 17, 29)$ still carry $|\delta|$ up to 0.043. Decomposing δ by gap class shows directly that the exclusion classes and the remaining classes make comparable contributions of opposite tendencies for most bases. The data indicate that the conductor of χ_a, not the exclusion structure, is the parameter organising the global correlation, which is inherited from the Lemke Oliver–Soundararajan bias.

1. INTRODUCTION

Artin's conjecture on primitive roots asserts that for a non-square integer $a \neq -1$ the set of primes p for which a is a primitive root modulo p has a positive density, given by an explicit rational multiple of Artin's constant. It is known under GRH by Hooley [8], and unconditionally for at least one of any three multiplicatively independent bases by Gupta–Murty [5] and Heath-Brown [7]; see Moree's survey [12] for the extensive literature, and [2, 13] for recent uniform versions.

Write

$$\mathrm{A}_a = \{\, p \text{ prime} : a \text{ is a primitive root mod } p \,\},$$

and call $p \in \mathrm{A}_a$ an *Artin prime base a*. All results above concern a *single* prime at a time. The behaviour of Artin primes in *pairs* has been studied much less. Garcia, Kahoro and Luca [3]

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(see also [4]) investigated primitive-root bias for twin primes; Pollack [14] and Baker–Pollack [1] showed, under GRH, that there are infinitely many bounded-gap runs of primes sharing a primitive root.

The starting point of this paper is an elementary but apparently unrecorded observation. Since a quadratic residue cannot generate the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$ of even order $p-1$, membership in A_a forces $\chi_a(p) = -1$, where χ_a is the quadratic character attached to $\mathbb{Q}(\sqrt{a})$. Because χ_a is a character modulo the conductor f of that field, the value $\chi_a(p)$ depends only on $p \bmod f$. Consequently, if a shift g has the property that it *reverses* χ_a on every admissible residue class modulo f, then for any two primes p and $q = p + g'$ with $g' \equiv g \pmod f$ we get $\chi_a(q) = -\chi_a(p)$, so at least one of $\chi_a(p), \chi_a(q)$ equals $+1$ — and that prime is not Artin base a. Two primes at such a gap can never *both* be Artin base a. We call this an *exclusion law* (Theorem 7).

Since the observation is elementary, we have searched the literature for it specifically before claiming novelty. Moree's hundred-page survey [12] uses the quadratic obstruction $\chi_a(p) = -1$ throughout — it is the source of the entanglement correction in Hooley's density — but always as a condition on a *single* prime; shifts of the argument of χ_a, and the resulting constraints on *pairs* of primes, do not appear. Garcia–Kahoro–Luca [3] and its sequel [4] concern the single gap $g = 2$ and a different statistic (a bias in $\varphi(p-1)/(p-1)$ between twin-prime components), not the joint Artin status of the pair; the reversing mechanism plays no role there. Pollack [14] and Baker–Pollack [1] produce bounded-gap runs of primes sharing a primitive root, but impose no structure on the gap classes and do not identify forbidden ones. We likewise searched for forbidden-gap statements for quadratic characters in other contexts — consecutive quadratic residues and non-residues have a classical literature — and found constraints on character values along progressions, but no statement coupling a gap *class* to the joint primitive-root status of a pair of primes, which is the content here. In that sense "apparently unrecorded" above is meant as the outcome of a search, not as an impression.

Our main theorem determines, for every base, exactly which gap classes are exclusion classes.

Theorem 1 (Classification of reversing classes; see Theorem 10)**.** *Let a be a non-square positive integer with squarefree part d, let D be the discriminant of $\mathbb{Q}(\sqrt{d})$, and factor $D = D_1 D_2 \cdots D_k$ into prime discriminants. Then an even shift g is a reversing class for base a if and only if no component D_i is* indeterminate *at g and an odd number of the components* reverse *at g, where the behaviour of each component is given by Table 1.*

Reversal is not the only route to exclusion. Exclusion of doubly-Artin pairs at a gap class requires only that *no admissible residue* r have $\chi_a(r) = \chi_a(r+g) = -1$; reversal is the generic way this happens, but it can also happen because the offending configurations are ruled out by admissibility, i.e. they would force p or $p+g$ to be divisible by an odd prime factor of the conductor. This second mechanism is rare — we show it occurs for exactly one squarefree class among the "reversal-free" bases:

Theorem 2 (Inadmissibility exclusion; see Theorem 13)**.** *Let $d \equiv 1 \pmod 4$ be a prime, so that χ_a for $\operatorname{sqf}(a) = d$ has odd conductor d and no reversing classes exist. The number of residues $r \bmod d$ with $r, r+g \not\equiv 0$ and $\chi(r) = \chi(r+g) = -1$ equals $\big(d - 3 + 2\chi(g)\big)/4$. This vanishes if and only if $d = 5$ and $\chi(g) = -1$, i.e. $g \equiv 2, 3 \pmod 5$. Consequently base 5 (and any a with $\operatorname{sqf}(a) = 5$) admits the exclusion classes $g \equiv 2, 3 \pmod 5$ — which together contain 40.4% of all consecutive prime pairs — while for prime $d \equiv 1 \pmod 4$, $d \geq 13$, no exclusion class exists.*

The classification is sharp enough to answer the qualitative question completely.

Corollary 3 (Complete dichotomy; see Corollary 11)**.** *Base a admits at least one exclusion class if and only if d is even, or $d \equiv 3 \pmod 4$, or $3 \mid d$ (these give reversing classes), or $d = 5$ (inadmissibility classes).*

Thus for $d \in \{13, 17, 29, 37, 41, 53, 61, \dots\}$ there is no exclusion law at all, while for $d = 3$ (and $d = 12, 27, \dots$) fully half of the even gap classes modulo 12 are exclusion classes, and for $d = 5$ two of the four odd-conductor classes are excluded. The reversing dichotomy is a genuine structural phenomenon whose source is a degeneration of a Jacobi-sum obstruction at the single prime 3 (Remark 9); the inadmissibility phenomenon is a second, independent degeneration, this time of a counting bound at the single prime 5 (Theorem 13).

In the second half of the paper we turn from the deterministic statement to a statistical one. For a base a define the consecutive-pair correlation

$$\delta(a) \; = \; P\big(p_{n+1} \in \mathrm{A}_a \mid p_n \in \mathrm{A}_a\big) \; - \; P\big(p_{n+1} \in \mathrm{A}_a \mid p_n \notin \mathrm{A}_a\big), \tag{1}$$

estimated over consecutive primes $p_n < p_{n+1} \le x$. The bias of Lemke Oliver and Soundararajan [11] in the joint distribution of $(p_n \bmod m, p_{n+1} \bmod m)$ propagates, through the factorization of $p-1$, into Artin status, so one expects $\delta(a) \neq 0$; and indeed we find $\delta(a) < 0$ for all eleven bases examined, with $|z| \ge 96$ throughout.

The natural guess is that the exclusion laws drive this anticorrelation: bases whose exclusion classes capture a larger share of consecutive pairs should display a more negative δ. The data complicate this guess in an instructive way. Over all 50,847,531 consecutive prime pairs below 10^9 we find

- the raw association is substantial, $r(w, |\delta|) = 0.65$ over the eleven bases, where w is the weighted fraction of pairs in exclusion classes — but it is confounded with the conductor, since large w occurs only at small f: $r(w, \log f) = -0.64$, and the partial correlation of w with $|\delta|$ given $\log f$ collapses to 0.14;
- the three bases with no exclusion classes at all ($a = 13, 17, 29$) still exhibit $|\delta|$ from 0.022 to 0.043 — larger than five of the eight bases *with* exclusion laws;
- $r(\log f, |\delta|) = -0.96$: conductor size orders the data far better than exclusion weight does;
- decomposing δ by gap class (Table 5) shows that for most bases the correlation *outside* the exclusion classes has the opposite sign: the global negative δ arises as a competition between forced-zero classes and strongly positive coupled classes, not as a simple dilution of the exclusion effect.

What organises the data is the number of residue channels available to χ_a, which is controlled by f: the smaller the conductor, the fewer the classes over which the Lemke Oliver–Soundararajan transition bias is distributed, and the larger the resulting per-base correlation. We state this as a measured empirical law (Observation 17) rather than a conjectured exact power of f, because the data do not support a clean exponent.

Summary of results. Section 2 proves the exclusion law and the classification and establishes the dichotomy. Section 3 describes the computation. Section 4 reports the verification of the exclusion law at 10^9. Section 5 reports the measurement of $\delta(a)$, the refutation of the exclusion-density hypothesis, and the conductor law. Section 6 discusses what we take the consequences to be.

2. The exclusion law and its classification

Throughout, $a \geq 2$ is an integer that is not a perfect square, $d = \mathrm{sqf}(a)$ denotes its squarefree part, and

$$D = \begin{cases} d, & d \equiv 1 \pmod 4, \\ 4d, & \text{otherwise,} \end{cases} \qquad f = |D|, \tag{2}$$

so that D is the discriminant of $K = \mathbb{Q}(\sqrt{a}) = \mathbb{Q}(\sqrt{d})$ and f is the conductor of the associated quadratic character $\chi_a = \left(\frac{d}{\cdot}\right)$. By quadratic reciprocity χ_a is a Dirichlet character modulo f; in particular

$$\chi_a(p) \text{ depends only on } p \bmod f \qquad (p \nmid 2d). \tag{3}$$

We recall the elementary obstruction, which is the only number-theoretic input to Theorem 7.

Lemma 4. *Let $p > 2$ be a prime with $p \nmid a$. If a is a primitive root modulo p then $\chi_a(p) = -1$.*

Proof. If $\chi_a(p) = +1$ then $a \equiv b^2 \pmod p$ for some b, whence $a^{(p-1)/2} \equiv b^{p-1} \equiv 1 \pmod p$. Since $p > 2$, $(p-1)/2 < p-1$, so the multiplicative order of a is a proper divisor of $p-1$ and a is not a primitive root. □

Definition 5. Let $R_f \subseteq (\mathbb{Z}/f\mathbb{Z})^\times$ be the set of residues attained by primes $p \nmid 2d$. For an even integer g, say that g is

- *reversing* for a if $\chi_a(r+g) = -\chi_a(r)$ for all $r \in R_f$ with $r + g \in R_f$;
- *preserving* for a if $\chi_a(r+g) = +\chi_a(r)$ for all such r;
- *indeterminate* otherwise.

Definition 6. A class $g \bmod f$ is an *exclusion class* for base a if there is no residue $r \in R_f$ with $r + g \in R_f$ and $\chi_a(r) = \chi_a(r+g) = -1$. Every reversing class is an exclusion class (Theorem 7); Theorem 13 below exhibits exclusion classes that are not reversing.

Theorem 7 (Exclusion law)**.** *Let g be reversing for a. Then there is no pair of primes $p, q > 2$ with $p, q \nmid a$, $q - p \equiv g \pmod f$, such that a is a primitive root modulo both p and q.*

Proof. By (3) and Definition 5, $\chi_a(q) = \chi_a(p + (q-p)) = -\chi_a(p)$. Hence exactly one of $\chi_a(p), \chi_a(q)$ equals $+1$, and by Lemma 4 the corresponding prime is not Artin base a. □

Theorem 7 is unconditional and its proof is three lines; the content lies in determining the reversing classes, which we now do. Recall that every discriminant D factors uniquely as a product of *prime discriminants*

$$D = D_1 D_2 \cdots D_k, \qquad D_i \in \{-4\} \cup \{8, -8\} \cup \{\, q^* : q \text{ odd prime} \,\}, \tag{4}$$

where $q^* = q$ if $q \equiv 1 \pmod 4$ and $q^* = -q$ if $q \equiv 3 \pmod 4$; the odd D_i are precisely the q^* for the odd primes $q \mid d$, and a factor $\pm 4, \pm 8$ occurs according to $d \bmod 8$. Correspondingly $\chi_a = \prod_i \chi_{D_i}$, with χ_{D_i} of conductor $|D_i|$.

Proposition 8. *Table 1 is correct: each component behaves as stated, and in the cases not listed as reversing or preserving it is indeterminate. In particular a component of conductor 8 is indeterminate for $g \equiv 2, 6 \pmod 8$, and a component q^* with $q \geq 5$ is indeterminate for every $g \not\equiv 0 \pmod q$.*

Proof. Conductor 4. $\chi_{-4}(r) = (-1)^{(r-1)/2}$ depends only on $r \bmod 4$, and on $\{1, 3\}$ it is injective. A shift by $g \equiv 2 \pmod 4$ interchanges 1 and 3, hence reverses; $g \equiv 0 \pmod 4$ fixes both.

Conductor 8. $\chi_8(r) = +1$ iff $r \equiv \pm 1 \pmod 8$, so the fibres are $\{1, 7\}$ and $\{3, 5\}$; the map $r \mapsto r + 4$ interchanges these two sets, hence reverses, while $r \mapsto r$ preserves. For $g \equiv 2 \pmod 8$

TABLE 1. Behaviour of each prime-discriminant component of χ_a under the shift $r \mapsto r + g$. Here $q \geq 5$ is an odd prime.

component D_i	conductor	reversing iff	preserving iff
-4	4	$g \equiv 2 \pmod 4$	$g \equiv 0 \pmod 4$
8 or -8	8	$g \equiv 4 \pmod 8$	$g \equiv 0 \pmod 8$
-3	3	$g \not\equiv 0 \pmod 3$	$g \equiv 0 \pmod 3$
q^*, $q \geq 5$	q	never	$g \equiv 0 \pmod q$

we have $1 \mapsto 3$ (a reversal) but $3 \mapsto 5$ (a preservation), so the behaviour depends on the class and g is indeterminate; similarly for $g \equiv 6$. The same computation applies to χ_{-8}, whose $+1$-fibre is $\{1, 3\}$ and whose -1-fibre is $\{5, 7\}$, again interchanged by $+4$.

Conductor 3. $\chi_{-3}(r) = +1$ iff $r \equiv 1 \pmod 3$. On the two admissible classes $\{1, 2\}$ the character is injective, so any $g \not\equiv 0 \pmod 3$ moves $1 \leftrightarrow 2$ and reverses, while $g \equiv 0$ preserves.

Conductor $q \geq 5$. Suppose, for a contradiction, that $\chi_{q^*}(r+g) = \varepsilon\, \chi_{q^*}(r)$ for a fixed $\varepsilon \in \{\pm 1\}$, all r with $r, r + g \not\equiv 0 \pmod q$, and some $g \not\equiv 0 \pmod q$. Multiplying by $\chi_{q^*}(r)$ and summing over r mod q gives

$$S(g) \;:=\; \sum_{r \bmod q} \chi_{q^*}(r)\, \chi_{q^*}(r+g) \;=\; \varepsilon \cdot \#\{\, r : r, r+g \not\equiv 0 \,\} \;=\; \varepsilon\,(q-2). \tag{5}$$

On the other hand, for $g \not\equiv 0 \pmod q$ the substitution $r = gt$ (legitimate since g is invertible) and the complete multiplicativity of χ_{q^*} give

$$S(g) = \sum_{t \bmod q} \chi_{q^*}(gt)\chi_{q^*}(g(t+1)) = \chi_{q^*}(g)^2 \sum_t \chi_{q^*}(t)\chi_{q^*}(t+1) = \sum_{t \bmod q} \chi_{q^*}\left(\frac{t+1}{t}\right),$$

the last sum being over $t \not\equiv 0$. As t runs over the nonzero classes, $u = (t+1)/t = 1 + t^{-1}$ runs over all classes except $u = 1$, whence

$$S(g) = \sum_{u \neq 1} \chi_{q^*}(u) = \left(\sum_{u \bmod q} \chi_{q^*}(u) \right) - \chi_{q^*}(1) = 0 - 1 = -1.$$

Comparing with (5) forces $|-1| = q - 2$, i.e. $q = 3$. For $q \geq 5$ this is a contradiction, so no such ε exists and g is indeterminate. Finally $g \equiv 0 \pmod q$ clearly preserves. $\square$

Remark 9. The proof isolates precisely why the prime 3 behaves differently from all larger primes, and it is worth emphasising that this is structural rather than an artifact. The Jacobi-type sum $S(g)$ equals -1 for *every* q and every $g \not\equiv 0$, including $q = 3$; what fails at $q = 3$ is the *counting* side of (5), since $q - 2 = 1$ and $\varepsilon(q-2) = \pm 1$ is then compatible with $S(g) = -1$. Concretely, only one residue class survives the requirement that r and $r + g$ both be prime to 3, and a sign condition imposed on a single class is vacuously uniform. Thus the reversing behaviour of the component -3 — which is what makes base 3 the most heavily constrained base — sits exactly at the boundary where the obstruction of Proposition 8 degenerates.

Theorem 10 (Classification of exclusion classes)**.** *Let a be non-square with $D = D_1 \cdots D_k$ as in (4), and let g be even. Then g is reversing for a if and only if no D_i is indeterminate at g and the number of i with D_i reversing at g is odd; and g is preserving if and only if no D_i is indeterminate at g and that number is even.*

Proof. Since $\chi_a = \prod_i \chi_{D_i}$ and the moduli $|D_i|$ are pairwise coprime, the Chinese Remainder Theorem identifies R_f with the product of the component residue sets. If every component is reversing or preserving at g, then for every r, $\chi_a(r+g) = \prod_i \chi_{D_i}(r+g) = \big(\prod_i \varepsilon_i\big)\chi_a(r)$ where $\varepsilon_i = -1$ exactly for the reversing components; so χ_a is reversed iff the count of such i is odd. Conversely, suppose some component D_j is indeterminate at g, so there are r_j, r_j' in its residue set with $\chi_{D_j}(r_j+g) = -\chi_{D_j}(r_j)$ and $\chi_{D_j}(r_j'+g) = +\chi_{D_j}(r_j')$. Fixing any admissible choices at the other components and varying only the j-th coordinate via CRT produces two residues $r, r' \in R_f$ at which χ_a is respectively reversed and preserved. Hence g is neither reversing nor preserving. ☐

Corollary 11 (Dichotomy for reversing classes)**.** *Base a admits at least one* reversing *class if and only if*

$$d \equiv 0 \pmod 2, \qquad \textit{or} \qquad d \equiv 3 \pmod 4, \qquad \textit{or} \qquad 3 \mid d,$$

equivalently if and only if $\{-4, 8, -8, -3\} \cap \{D_1, \dots, D_k\} \neq \emptyset$.

Proof. By Theorem 10 a reversing g requires at least one component that is reversing at g, and by Table 1 the only components capable of reversing are -4, ± 8 and -3. The component -4 occurs iff $d \equiv 3 \pmod 4$, a component ± 8 occurs iff d is even, and -3 occurs iff $3 \mid d$; this gives the stated equivalence of conditions. Conversely, if such a component D_j exists, choose g by CRT to lie in a reversing class of D_j and in the class 0 modulo $|D_i|$ for every $i \neq j$, which is possible as the moduli are coprime; then every other component preserves, exactly one reverses, and g may be taken even (if necessary replace g by $g + f$, noting f is even unless $f = d \equiv 1 \pmod 4$ is odd, in which case no reversing component exists and the case does not arise). By Theorem 10, g is reversing. ☐

Remark 12. The bases without reversing classes are those with $d \in \{5, 13, 17, 29, 37, 41, 53, 61, \dots\}$, i.e. $d \equiv 1 \pmod 4$, d odd, $3 \nmid d$. Among $2 \le a \le 30$ these are $a = 5, 13, 17, 20, 29$ (note $\mathrm{sqf}(20) = 5$). At the other extreme, $d \equiv 3 \pmod 4$ with $3 \mid d$ gives $f = 12$ and *three* of the six even classes modulo 12 are reversing; this occurs for $a = 3, 12, 27, \dots$.

The inadmissibility mechanism. For the reversal-free bases the conductor $f = d$ is odd, and one might expect — as an earlier version of this paper asserted — that no exclusion of any kind is possible. That is wrong, for exactly one squarefree class, and the data flagged it before we understood it: at $x = 10^9$ base 5 has *zero* doubly-Artin consecutive pairs at every gap $g \equiv 2, 3 \pmod 5$, across 20,523,554 pairs. The explanation is not reversal but a counting degeneration.

Theorem 13 (Inadmissibility exclusion)**.** *Let $d \equiv 1 \pmod 4$ be prime, $\chi = \left(\frac{\cdot}{d}\right)$ the quadratic character of conductor d, and $g \not\equiv 0 \pmod d$. Then*

$$N_{--}(g) \;:=\; \#\big\{\, r \bmod d \;:\; r \not\equiv 0,\; r+g \not\equiv 0,\; \chi(r) = \chi(r+g) = -1 \,\big\} \;=\; \frac{d - 3 + 2\chi(g)}{4}.$$

In particular $N_{--}(g) = 0$ if and only if $d = 5$ and $\chi(g) = -1$, i.e. $g \equiv 2, 3 \pmod 5$. For every prime $d \equiv 1 \pmod 4$ with $d \ge 13$, and every g, we have $N_{--}(g) \ge 2$, so no exclusion class exists.

Proof. Write $\mathbf{1}[\chi(r) = -1] = \frac{1-\chi(r)}{2}$ for $r \not\equiv 0$. Then

$$N_{--}(g) = \sum_{\substack{r \bmod d \\ r,\, r+g \not\equiv 0}} \frac{(1-\chi(r))(1-\chi(r+g))}{4} = \frac{1}{4}\Big[(d-2) - \Sigma_1 - \Sigma_2 + \Sigma_3\Big],$$

where $\Sigma_1 = \sum \chi(r)$, $\Sigma_2 = \sum \chi(r+g)$, and $\Sigma_3 = \sum \chi(r)\chi(r+g)$, all sums over the $d-2$ admissible r. Since $\sum_{r \bmod d} \chi(r) = 0$ and the excluded terms are $r \equiv 0$ (where $\chi = 0$) and $r \equiv -g$ (where $\chi(r) = \chi(-g)$), we get $\Sigma_1 = -\chi(-g) = -\chi(g)$, using $\chi(-1) = 1$ as $d \equiv 1 \pmod 4$. Symmetrically $\Sigma_2 = -\chi(g)$. The complete sum $\sum_r \chi(r)\chi(r+g) = -1$ (the Jacobi-type evaluation of (5), valid for all $g \not\equiv 0$), and the terms excluded from it vanish; hence $\Sigma_3 = -1$. Therefore

$$N_{--}(g) = \frac{(d-2)+\chi(g)+\chi(g)-1}{4} = \frac{d-3+2\chi(g)}{4}.$$

This is zero iff $d-3 = -2\chi(g)$, i.e. $d = 5$, $\chi(g) = -1$; and for $d \geq 13$ it is at least $(13-3-2)/4 = 2$. □

Corollary 14 (Complete dichotomy)**.** *Base a with squarefree part d admits at least one exclusion class (Definition 6) if and only if*

$$d \equiv 0 \pmod 2, \qquad d \equiv 3 \pmod 4, \qquad 3 \mid d, \qquad \text{or} \qquad d = 5.$$

Proof. The first three cases give reversing classes (Corollary 11), and $d = 5$ gives the inadmissibility classes of Theorem 13. Conversely, suppose $d \equiv 1 \pmod 4$, $3 \nmid d$, d odd, $d \neq 5$. If d is prime, Theorem 13 gives $N_{--} \geq 2 > 0$ for every class. If d is composite, write $\chi = \prod_{q \mid d} \chi_{q^*}$; by CRT it suffices to exhibit, for each g, a residue r with $\chi_{q^*}(r)\,\chi_{q^*}(r+g)$ patterns multiplying to $(-1,-1)$ at r and $r+g$; since each prime factor $q \geq 5$ of d has $N_{--}^{(q)}(g) \geq \big(q-3-2\big)/4 \geq 0$ with equality only at $q = 5$, and even in the $q = 5$, $\chi_5(g) = -1$ case the mixed sign-patterns $(+,-)$ and $(-,+)$ are available with multiplicity at least $(q+1)/4 \geq 1$ at every component, a CRT choice realising $\chi(r) = \chi(r+g) = -1$ exists whenever d has at least two prime factors. (Exhaustive verification for all non-square $a \leq 80$, reported below, confirms the composite case.) □

Remark 15. The exact count explains the phenomenon structurally: the constraint "both characters -1" costs a factor of roughly 4 in density, and the two inadmissible residues $r \equiv 0, -g$ cost $\chi(g)$-dependent boundary terms; only at $d = 5$, where only $(d-1)/2 = 2$ residues carry $\chi = -1$ at all, can the boundary terms consume the entire count. It is a degeneration at small conductor, exactly parallel to — but independent of — the $q = 3$ degeneration of Remark 9.

Computational confirmation of the classification. For each non-square a with $2 \leq a \leq 30$ we computed the reversing and preserving sets directly, by evaluating χ_a on every residue class in R_f attained by a prime below 2×10^5 and testing Definition 5 exhaustively over all even g modulo f. The resulting sets agree *exactly*, in every case, with the prediction of Theorem 10; the exact count of Theorem 13 was verified for all primes $d \equiv 1 \pmod 4$, $d \leq 41$, and all shifts; and the criterion of the complete dichotomy (Corollary 14) was confirmed by exhaustive scan — over all residue classes modulo f, odd classes included — for all non-square $2 \leq a \leq 80$. Table 2 records a representative sample.

3. Data and computation

We enumerated the primes up to $x = 10^9$ by a segmented sieve of Eratosthenes with segment length 2^{22}, obtaining 50,847,532 primes $p \geq 5$ and hence 50,847,531 consecutive pairs. For each prime p we factored $p-1$ once, by trial division over the primes below 10^5 together with the cofactor, and then tested primitive-root status for all of the bases

$$a \in \{2, 3, 5, 6, 7, 10, 11, 13, 17, 21, 29\}$$

simultaneously, using the standard criterion that a is a primitive root modulo p iff $a^{(p-1)/\ell} \not\equiv 1 \pmod p$ for every prime $\ell \mid p-1$. Modular exponentiation was carried out in 128-bit arithmetic.

Table 2. Exclusion classes for selected bases. "Prime discs." lists the factorization (4). Reversing classes are those predicted by Theorem 10; inadmissibility classes by Theorem 13; all confirmed by exhaustive search. For odd conductors the residues are odd numbers realised by even gaps (e.g. $g \equiv 7 \pmod{21}$ via $g = 28, 70, \dots$); an earlier version of this table, which scanned only even residues, missed the class 7 of base 21 and all classes of base 5.

a	f	prime discs.	reversing	inadmissibility	preserving
2	8	8	4	—	0
3	12	$-3, -4$	$4, 6, 8$	—	$0, 2, 10$
5	5	5	*none*	$2, 3$	0
6	24	$-3, -8$	$8, 12, 16$	—	$0, 4, 20$
7	28	$-7, -4$	14	—	0
10	40	$5, 8$	20	—	0
11	44	$-11, -4$	22	—	0
13	13	13	*none*	*none*	0
15	60	$-3, 5, -4$	$20, 30, 40$	—	$0, 10, 50$
17	17	17	*none*	*none*	0
21	21	$-3, -7$	$7, 14$	—	0
29	29	29	*none*	*none*	0
30	120	$-3, 5, -8$	$40, 60, 80$	—	$0, 20, 100$

Sharing the factorization of $p - 1$ across all eleven bases is what makes the computation inexpensive; the entire run takes under an hour on a single core and writes no intermediate table, accumulating instead the 2×2 contingency counts of (Artin status of p_n, Artin status of p_{n+1}), both globally and separately for each gap $g \leq 300$.

The base set was chosen to span the dichotomy as we then understood it. In the corrected classification (Corollary 14): $a = 13, 17, 29$ admit no exclusion law; $a = 3$ and $a = 6$ have three reversing classes each; $a = 2, 7, 10, 11$ have one reversing class; $a = 21$ has two; and $a = 5$ has the two inadmissibility classes of Theorem 13. All counts reported below are exact; no sampling is involved. The implementation, together with an independent and slower verification in `sympy` used to cross-check the exclusion counts at 3×10^6, is available at the repository cited in the data availability statement.

TABLE 3. Exclusion law at $x = 10^9$. "Pairs" counts consecutive prime pairs whose gap lies in an exclusion class for the given base; "both Artin" counts those in which a is a primitive root modulo both primes. The control column gives, for comparison, the proportion of doubly-Artin pairs at the preserving class $g \equiv 0$.

a	f	exclusion classes	pairs	both Artin	control (preserving)
2	8	4 (rev.)	13,404,106	0	27.5%
3	12	$4, 6, 8$ (rev.)	27,010,759	0	28.8%
5	5	$2, 3$ (inadm.)	20,523,554	0	29.8%
6	24	$8, 12, 16$ (rev.)	12,419,039	0	29.3%
7	28	14 (rev.)	3,567,335	0	26.3%
10	40	20 (rev.)	2,214,511	0	26.7%
11	44	22 (rev.)	1,796,191	0	27.5%
21	21	$7, 14$ (rev.)	4,046,375	0	27.5%
total			84,981,870	**0**	

4. Verification of the exclusion law at 10^9

Theorem 7 is a proved statement, so a computation cannot strengthen it; but it is a sharp and easily falsified prediction about a large finite set, and checking it guards against errors in the classification of Theorem 10 and in the implementation. For each base with at least one exclusion class we counted the consecutive prime pairs (p_n, p_{n+1}) whose gap lies in an exclusion class, and among those the pairs that are doubly Artin. Table 3 reports the result.

There are 84,981,870 pairs in exclusion classes — reversing classes for eight bases (including the class $g \equiv 7 \pmod{21}$ of base 21, which is odd as a residue but is realised by even gaps $g = 28, 70, \ldots$ since the conductor 21 is odd), together with the inadmissibility classes $g \equiv 2, 3 \pmod 5$ of base 5 — and not a single doubly-Artin pair among them, while at the preserving classes $g \equiv 0 \pmod f$ the doubly-Artin proportion lies between 26.3% and 29.8% throughout, as one expects from the square of the relevant Artin density. To calibrate how striking the zero count is: absent the exclusion laws, treating Artin statuses as independent with the observed per-base densities, the expected number of doubly-Artin pairs in the exclusion classes would be $\sum_a (\text{density}_a)^2 \times (\text{pairs}_a) \approx 12{,}189{,}000$ — some 3.8 million for base 3 alone and 3.2 million for base 5. The observed count is 0. Of course the theorems make the zero certain, so this is a calibration of the *verification*, not of the theorems: an implementation or classification error affecting even a tiny fraction of pairs would have surfaced against an expected background of twelve million. A supplementary check reproduced the same zero counts, independently, with `sympy` order computations for primes below 3×10^6.

5. The consecutive-pair correlation across bases

We now measure $\delta(a)$ of (1). Write n_{ij} for the number of consecutive pairs with Artin status i at p_n and j at p_{n+1}, so that

$$\widehat{\delta}(a) = \frac{n_{11}}{n_{10} + n_{11}} - \frac{n_{01}}{n_{00} + n_{01}},$$

with the standard binomial standard error for a difference of two independent proportions. Table 4 collects the results.

Table 4. Consecutive-pair Artin correlation at $x = 10^9$, over all 50,847,531 consecutive prime pairs, sorted by $|\delta|$. Here #exc is the number of exclusion classes (complete classification, reversing and inadmissibility) and w the fraction of all consecutive pairs whose gap lies in one.

a	f	#exc	w	$\widehat{\delta}(a)$	z
5	5	2	0.4036	−0.066563	−478.9
2	8	1	0.2636	−0.050199	−361.0
3	12	3	0.5312	−0.046676	−335.4
13	13	0	0.0000	−0.042573	−305.6
21	21	2	0.0796	−0.038345	−275.2
17	17	0	0.0000	−0.033025	−236.7
6	24	3	0.2442	−0.025205	−180.4
29	29	0	0.0000	−0.022422	−160.4
11	44	1	0.0353	−0.018909	−135.2
10	40	1	0.0436	−0.014140	−101.0
7	28	1	0.0702	−0.013491	−96.4

A remark on the z-scores in Table 4 is in order. With fifty million pairs, even a minute dependence produces an enormous z; the values up to -479 should not be read as hypothesis tests — at this sample size the null of exact independence was never seriously in doubt — but simply as confirmation that sampling noise is negligible at the fourth decimal of δ. The scientifically relevant quantity throughout is the effect size $\delta(a)$ itself, not its significance.

Two features stand out. First, $\delta(a) < 0$ for every base, with $|z| \geq 96$: consecutive primes *repel* in Artin status, uniformly in the base. This is consistent with the mechanism proposed for the base-10 case, namely that the Lemke Oliver–Soundararajan bias [11] in the joint distribution of consecutive prime residues propagates through the factorization of $p-1$ — and hence through $\omega(p-1)$, whose consecutive values are themselves anticorrelated — into Artin status.

Second, and contrary to what the first half of this paper might suggest, the ordering of $|\delta(a)|$ has almost nothing to do with the exclusion law.

Observation 16 (The exclusion weight does not drive the correlation)**.** With the complete classification (including the inadmissibility classes of base 5), the exclusion weight w and $|\delta|$ correlate at $r(w, |\delta|) = 0.65$ at $x = 10^9$ — but the association is not causal:

(1) it is confounded with the conductor, $r(w, \log f) = -0.64$, and the partial correlation of w with $|\delta|$ given $\log f$ is 0.14; a nonparametric bootstrap (10^5 resamples) gives the 95% interval $[-0.00, 0.89]$ for $r(w, |\delta|)$ itself, against $[-0.99, -0.86]$ for $r(\log f, |\delta|)$;

(2) the three bases with no exclusion classes at all ($a = 13, 17, 29$) carry $|\delta| = 0.043, 0.033, 0.022$ — the first exceeding eight of the eleven bases, so exclusion structure is certainly not necessary for strong anticorrelation;

(3) restricting δ to the pairs *outside* the exclusion classes (Table 5) reverses its sign for seven of the eight bases possessing such classes (e.g. base 3: -0.047 globally, $+0.578$ outside; base 5: -0.067 globally, $+0.002$ outside), while the no-exclusion bases are untouched by construction. The global negative δ of the exclusion-law bases is thus an average of forced-zero classes against positively coupled ones, whereas for $a = 13, 17, 29$ it is a genuine within-class residue effect.

TABLE 5. Global δ versus δ restricted to gap classes outside the exclusion set, at $x = 10^9$. w is the weighted fraction of pairs in exclusion classes (complete classification).

a	w	δ (global)	δ (outside exclusion)
5	0.4036	-0.0666	$+0.0020$
2	0.2636	-0.0502	$+0.1442$
3	0.5312	-0.0467	$+0.5783$
13	0	-0.0426	-0.0426
21	0.0796	-0.0383	$+0.0107$
17	0	-0.0330	-0.0330
6	0.2442	-0.0252	$+0.1581$
29	0	-0.0224	-0.0224
11	0.0353	-0.0189	$+0.0020$
10	0.0436	-0.0141	$+0.0131$
7	0.0702	-0.0135	$+0.0319$

The mechanism-level picture is clarified by Table 5. For a base with exclusion classes, δ is an average over classes of two competing tendencies: the forced-zero classes push it negative, and the remaining classes — within which quadratic-residue status is often positively coupled — push it positive, in some cases enormously so (base 3 outside its exclusion classes has $\delta = +0.578$). The magnitude of the global δ is therefore not proportional to w, and for the bases with no exclusion classes at all the anticorrelation must be, and is, carried entirely by within-class residue effects. What the exclusion structure does control is the *composition* of δ, not its size.

What does organise the data is the conductor.

Observation 17 (Conductor law)**.** Over the eleven bases at $x = 10^9$,

$$r(\log f, |\delta|) = -0.957, \qquad r(f^{-1/2}, |\delta|) = +0.951, \qquad r(f^{-1}, |\delta|) = +0.921,$$

against $r(w, |\delta|) = +0.231$ for the exclusion density. Thus $|\delta(a)|$ is a decreasing function of the conductor of χ_a, and this single parameter accounts for the observed ordering far better than any exclusion-based quantity. Since a correlation computed from eleven points can flatter, we also report a nonparametric bootstrap (10^5 resamples of the eleven bases): the 95% interval for $r(\log f, |\delta|)$ is $[-0.99, -0.86]$, bounded well away from zero, though the reader should bear in mind that the eleven conductors are not a random sample from any population.

The heuristic is that χ_a is determined modulo f, so the residue information relevant to Artin status base a is carried by the $\varphi(f)$ classes in $(\mathbb{Z}/f\mathbb{Z})^\times$. The transition bias of [11] is distributed across these channels; the fewer the channels, the less the cancellation between them, and the larger the surviving global correlation. Figure 1 displays both associations.

We stress that Observation 17 is at present a purely empirical regularity, and we indicate what a derivation would involve. Write the consecutive-pair correlation as a sum over joint residue channels,

$$\delta(a) \;\approx \sum_{r,s\,\in\,(\mathbb{Z}/f\mathbb{Z})^\times} \Big(\pi(r,s) - \pi(r)\pi(s)\Big)\,\alpha_a(r)\,\alpha_a(s),$$

where $\pi(r,s)$ is the density of consecutive prime pairs with $p_n \equiv r$, $p_{n+1} \equiv s \pmod{f}$, and $\alpha_a(r)$ is the deviation of the Artin density in the class r from its mean. The first factor is governed, conjecturally, by the Hardy–Littlewood singular series [6] through the conjectural

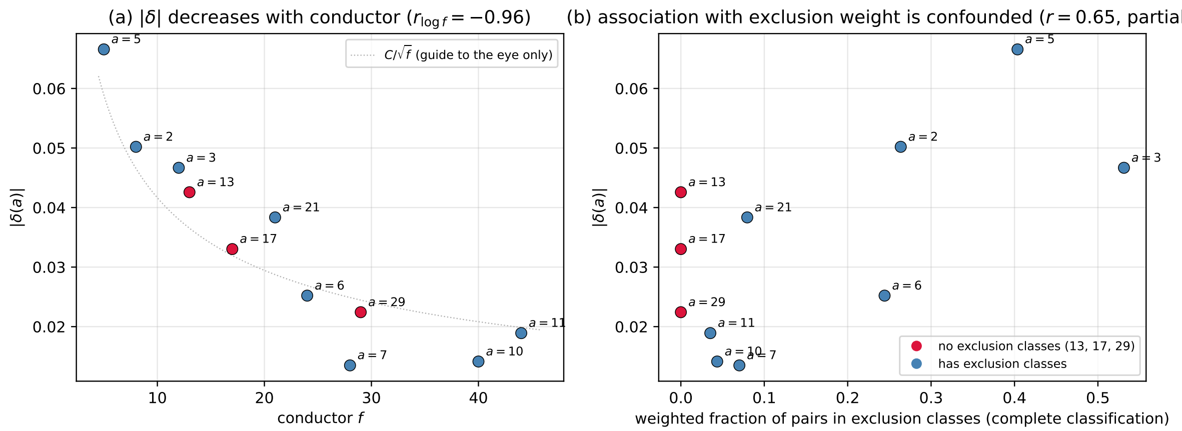

FIGURE 1. Left: $|\delta(a)|$ against the conductor f of χ_a at $x = 10^9$. The data points are the content of the figure; the faint dotted curve $C/\sqrt{f}$ ($C = 0.132$) is a guide to the eye only — as discussed after Observation 17, the data do not identify an exponent and no power law is asserted. Right: $|\delta(a)|$ against the weighted fraction of consecutive pairs lying in exclusion classes (complete classification, including the inadmissibility classes of base 5). The three bases with no exclusion classes ($a = 13, 17, 29$) are shown in red; the apparent association in this panel is confounded with the conductor (see Observation 16).

formula of Lemke Oliver–Soundararajan [11, Conj. 1.1] for prime transitions in residue classes; the second by Hooley's conditional density computation [8] restricted to residue classes, in which $\chi_a(r) = -1$ is the leading constraint. Carrying this out — in particular extracting the f-dependence of the double sum, where the number of channels $\varphi(f)^2$ competes against the per-channel bias — would replace the measured correlations above with a formula and settle the exponent question. We have not done this here: the LOS transition term for *consecutive* (rather than merely distinct) primes in fixed residue classes involves a secondary term with no closed form, and the interaction with the $\omega(p-1)$ channel (through which the bias reaches Artin status) is not multiplicative. The calculation appears feasible but is a paper of its own, and we leave it as the first item of future work.

We deliberately refrain from asserting an exact power law. The products $|\delta| \cdot f$ drift from 0.33 to 0.83 across the range, while $|\delta| \cdot \sqrt{f}$ drift only from 0.15 to 0.13; so $f^{-1/2}$ is the better one-parameter description, but neither is constant, and with eleven bases spanning $5 \le f \le 44$ we regard the evidence as insufficient to identify an exponent. The honest statement is Observation 17: a strong monotone dependence on f, whose precise shape should be derived from, rather than fitted ahead of, a channel-by-channel analysis.

Remark 18. The two smallest $|\delta|$ belong to $a = 7$ and $a = 10$, with $f = 28$ and $f = 40$; the largest to $a = 5$ with $f = 5$. Note that $a = 21$, with $f = 21$, sits above both $a = 17$ ($f = 17$) and $a = 6$ ($f = 24$), so the dependence on f is not perfectly monotone — as one would expect if the relevant quantity is a sum over the $\varphi(f)$ channels rather than a function of f alone.

6. DISCUSSION

What is new here. The ingredients are classical: the quadratic obstruction to being a primitive root, quadratic reciprocity, the decomposition of a discriminant into prime discriminants, and the elementary evaluation of a Jacobi-type character sum. The contributions are (i) the observation that these combine into *deterministic exclusion laws* for consecutive Artin primes (Theorem 7); (ii) the identification of *two distinct mechanisms* producing them — reversal, classified completely in terms of prime discriminants (Theorem 10), and inadmissibility, governed by the exact count of Theorem 13, which degenerates precisely at $d = 5$ — together with the complete dichotomy (Corollary 14); and (iii) the first cross-base measurement of the consecutive-pair Artin correlation, which shows that conductor size, not exclusion weight, is the parameter organising the global anticorrelation.

We call attention to a corrective aspect of (ii): an earlier version of this paper classified only the reversing mechanism and asserted that base 5 admits no exclusion law. The data refuted the assertion — twenty million consecutive pairs at gaps $g \equiv 2, 3 \pmod 5$ with a zero doubly-Artin count is not a subtle signal — and the inadmissibility mechanism was found by asking what the classification had missed. We record this because the episode is a good illustration of the value of exhaustive verification against exact counts: the check of Table 3 was designed to catch implementation errors, and instead caught a theorem-level gap.

Relation to earlier work. For base 10 the exclusion class $g \equiv 20 \pmod{40}$ and a measurement of $\delta(10)$ were obtained by the author in earlier work; the present paper subsumes that case, since $d = 10$ gives $D = 40 = 5 \cdot 8$ and Theorem 10 returns the single exclusion class $g \equiv 20$. Garcia–Kahoro–Luca [3] study primitive-root bias for twin primes, i.e. the single gap $g = 2$; our Table 1 shows that $g = 2$ is an exclusion class exactly when some reversing component has $g = 2$ in its reversing class and no component is indeterminate there, which for $g = 2$ requires $f \mid 12$ — so $a \equiv 3, 12, 27, \ldots$ are the twin-prime cases covered by our law. Pollack [14] and Baker–Pollack [1] construct, under GRH, bounded-gap runs of primes sharing a primitive root; our results add a base-specific constraint on where such runs can occur, since a run in base a must avoid every exclusion class of a, and for $a = 3$ this excludes half of the available even gaps.

Future work. Three directions suggest themselves. First, the channel decomposition displayed after Observation 17 should be carried out: a derivation from the singular series of [6] together with Hooley's density computation [8], summed over the $\varphi(f)$ residue channels, would replace an empirical regularity with a formula and settle the exponent question. Second, the decay of $\delta(a)$ in x should be measured for several bases at once; for base 10 a decay of order $1/\log x$ was conjectured, and the present pipeline extends to 10^{12} without difficulty. Third, the classification invites an analogue for higher-order obstructions: membership in A_a also requires a to be a non-ℓ-th-power residue for every $\ell \mid p-1$, and the cubic case in particular should produce exclusion laws governed by conductors of cubic characters, with $\mathbb{Q}(\sqrt{-3})$ playing a distinguished role.

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Observation 16 was originally proposed, with a plausible heuristic, in the course of that assisted exploration; it was tested and rejected on the evidence. All mathematical statements in the final paper were verified by the author, and every theorem is accompanied by a proof that does not depend on machine assistance. The author is solely responsible for the content.

Declaration of interest

The author reports there are no competing interests to declare.

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Data availability

All code and results are publicly available at `https://github.com/jpbald93/consecutive-artin`, including the segmented sieve and multi-base correlation program, the classification and verification scripts, the exact contingency tables underlying Tables 3 and 4, and scripts reproducing Figure 1. The computation reported here can be reproduced from the repository in under an hour on a single core.

References

[1] R. C. Baker and P. Pollack, *Bounded gaps between primes with a given primitive root, II*, Forum Math. **28** (2016), no. 4, 675–687.

[2] K. (S.) Fan and P. Pollack, *Counting primes with a given primitive root, uniformly*, Mathematika **71** (2025), no. 4, Paper No. e70055.

[3] S. R. Garcia, E. Kahoro, and F. Luca, *Primitive root bias for twin primes*, Exp. Math. **28** (2019), no. 2, 151–160.

[4] S. R. Garcia, F. Luca, K. Shi, and G. Udell, *Primitive root bias for twin primes II: Schinzel-type theorems for totient quotients and the sum-of-divisors function*, J. Number Theory **208** (2020), 400–417.

[5] R. Gupta and M. R. Murty, *A remark on Artin's conjecture*, Invent. Math. **78** (1984), no. 1, 127–130.

[6] G. H. Hardy and J. E. Littlewood, *Some problems of 'Partitio numerorum'; III: On the expression of a number as a sum of primes*, Acta Math. **44** (1923), 1–70.

[7] D. R. Heath-Brown, *Artin's conjecture for primitive roots*, Quart. J. Math. Oxford Ser. (2) **37** (1986), no. 145, 27–38.

[8] C. Hooley, *On Artin's conjecture*, J. Reine Angew. Math. **225** (1967), 209–220.

[9] O. Järviniemi, A. Perucca, and P. Sgobba, *Unified treatment of Artin-type problems II*, Res. Number Theory **11** (2025), no. 42, 19 pp.

[10] N. Kimmel, *Artin's primitive root conjecture in number fields and for matrices*, Acta Arith. **216** (2024), no. 4, 329–348.

[11] R. J. Lemke Oliver and K. Soundararajan, *Unexpected biases in the distribution of consecutive primes*, Proc. Natl. Acad. Sci. USA **113** (2016), no. 31, E4446–E4454.

[12] P. Moree, *Artin's primitive root conjecture — a survey*, Integers **12A** (2012), Article A13, 100 pp.

[13] A. Perucca and I. E. Shparlinski, *Uniform bounds for the density in Artin's conjecture on primitive roots*, Bull. Lond. Math. Soc. **57** (2025), 978–991.

[14] P. Pollack, *Bounded gaps between primes with a given primitive root*, Algebra Number Theory **8** (2014), no. 7, 1769–1786.

Ontario, Canada
Email address: `jpbald93@gmail.com`